

THE EMS MAPPING PROJECT

User Guidebook

V4

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1. Interface

UPDATED IN V4 — The two tabs are named differently.

The left edge of the screen contains two tabs and two other icons:

Feature/Button	What it does
Map Setup	Contains... - Location Settings, - Drawing tool for Areas and Categories, - Place to upload Incident CSV files - A Summary panel, - An option to Export session-specific files.
Statistical Analysis	Running statistical tests on your data.
Guidebook icon	Opens the guidebook PDF in a new tab.
Reset All icon	Located at the bottom of the sidebar. - Clears polygons, CSV rows, custom filters, and Custom Analysis Groups (Section 7). - Map center default and station location are preserved.

Resizable Sidebar

UPDATED — You can now drag the right edge of the sidebar to make it wider or narrower.

A thin drag handle sits on the sidebar's right border. Hold and drag it to resize the panel if column names or location lists are getting cut off. The map canvas will resize automatically.

2. Floating Map Controls: Display, Filters & Legend

UPDATED — View Options, Heatmap Mode, Data Filters, Custom Filters, and the Legend now float directly over the map as popouts that can be moved around the map.

When the site loads, you'll see a small toolbar labeled MAP near the top of the map canvas with two options: Display and Filters. Clicking one opens the corresponding floating panel; clicking the X collapses it back down to the toolbar. Every floating panel (Display, Filters, and the Legend) can be moved to anywhere on screen by its header.

Feature/Button	What it does
Display panel	Contains - The Areas/Both/Categories view-mode buttons - The Incident Volume vs. Efficiency (Distance/Time) heatmap options, - The Show Labels options.
Filters panel	Contains - Clear All Map Filters, - Auto-generated filters, - A Custom Filters section
Legend	Displays the corresponding color gradient scale depending on active heatmap mode.

NOTE: Panel positions are not saved between page loads and will reset to their default spots (Display top-left of the map, Filters below it, Legend bottom-center) whenever you refresh.

Switching the heatmap mode (Incident Volume vs. Efficiency) or the view mode (Areas/Both/Categories) from the Display panel automatically syncs with the Statistical Analysis tab (Section 7).

3. Map Center & Station

Feature/Button	What it does
Set Map Location	Prompts for an address/landmark/ZIP code and recenters the map when you confirm an auto-located result.
Set Station	Click the button, then click anywhere on the map to drop the station marker. Drag to fine-tune, right-click to delete.
Set as Default	Saves the current pan/zoom level as the map's new default view.

4. Drawing & Managing Areas and Categories

Click a section header to expand it.

Areas

Feature/Button	What it does
Start/Finish	Start, click points on the map to create vertices of a polygon around the desired area on the map. Finish, then name the polygon to match a value in your CSV's location column <u>exactly</u> (case-sensitive).
Clear Areas	Deletes all (non-category) existing areas. Categories are unaffected.

Categories

UPDATED — Category visibility is controlled from the Display floating panel's (Section 2) view-mode buttons (Areas/Both/Categories).

Feature/Button	What it does
Start / Finish	Requires a CSV to already be loaded. Clicking Start reveals a search box to check every location ID that belongs to this category Draw the polygon and click Finish to name it (or add it to an already existing category).
Clear Categories	Deletes all categories. Areas are unaffected.

While drawing a category, hovering the outline over existing areas highlights and auto-checks them in the location list.

Summary & Export Files

The Summary panel displays locations by highest count and efficiency ranking.

5. Editing an Existing Area or Category

UPDATED — This entire feature is new.

Click any area or category on the map to open its popup, then click the Edit Area (or Edit Category Part) button at the top of the popup. This switches the map into edit mode for that one polygon.

The Edit Toolbar

A small toolbar appears near the top-left of the map while editing:

Feature/Button	What it does
Save Changes	Saves any changes in vertex position, insertions, or deletions, and clears previous distance data (so efficiency numbers can be recalculated on the next use).
Rename	Change an Area/Category name. Renaming a category part renames every polygon piece that shares that category name.
Edit Locations	Categories only. Opens a searchable checklist of every known location ID so you can add or remove which locations you want fed into this category, without having to redraw it.
Cancel	Discards all changes made and restores the polygon's original shape.

Vertex Editing

Feature/Button	What it does
Drag a vertex	Click and hold the solid circle, drag to reposition, release.
Click a midpoint	Small hollow circles appear at the midpoint of each edge — click one to insert a new vertex there and move as necessary.
Right-click a vertex	Deletes that vertex (a polygon must keep at least 3 points).

Enter saves changes and Escape cancels them while the edit toolbar is open

6. Custom Filters

Pick a column, name the filter, choose Numeric Ranges or Categorical Groups depending on the nature of your data in that column, then define the appropriate ranges/groups.

UPDATED — Each saved custom filter card now has an Edit button next to the Delete option. Editing reopens the same popup with the filter's current column, name, type, and ranges/groups so you can adjust them without recreating the filter from scratch.

Logic for everything else remains the same as for standard filters. Ensure case-sensitive and exact name matching.

7. Statistical Analysis

UPDATED — This lets you run real significance tests on your incident data.

Switch to the Statistical Analysis tab.

Metric

Feature/Button	What it does
Incident Volume	Tests whether incident counts differ across the groups you select (a chi-square test).
Efficiency	Tests whether response efficiency (miles traveled per minute of response time) differs from the overall average for each selected group. Requires a CSV timeElapsed column, a set station location, and requires that distance data already be calculated: switch the map's Mode to Efficiency once if you haven't already to trigger the distance calculation.

Choosing a metric here automatically syncs the map's Heatmap Mode to match (Incident Volume → the incident-count heatmap, Efficiency → the distance/time heatmap), and vice versa.

Compare (Rows)

This defines what each row of your analysis represents. Click Source to choose what you're comparing:

Feature/Button	What it does
Locations/Areas*	Default setting. A viewing scope selector (Areas/Both/Categories) matching the map's view modes updates the map's Display.
Source	Options correspond to existing filterable CSV columns (including custom filters), and rows become each of the column's distinct values.
Custom Analysis Groups	Only appears once you've defined at least one group (see below), and group constituents become your named combination groups.

*When Scope = Both, categories in the list can be expanded to select specific individual member areas within a category, while selecting other categories in their entirety.

At least 2 items must be selected here to run any test.

Split by (Columns)

Only shown when Metric = Incident Volume (efficiency tests in this program are designed to always be one-dimensional). Works the same way as the Rows selector, with an extra option:

Feature/Button	What it does
None (Goodness-of-Fit)	Default. Runs a 1-dimensional test asking whether the incident counts are evenly distributed across the selected Row groups.
Locations/Areas, Column, Custom filter, Custom Analysis Groups	Same options as for Rows. Selecting one turns the test into a 2-dimensional test crossing Rows against Columns To run this, at least 2 items must be checked on this axis as well.

Custom Analysis Groups

UPDATED — A way to define comparison groups that combine multiple filter conditions.

Expand the Custom Analysis Groups sub-section to manage these:

Feature/Button	What it does
Add Group	Opens a popup. Give the group a name and add one or more conditions from incident data file column names and/or custom filters. All conditions must be true for an incident to be included in the group.
Auto-Generate Combinations	Pick two or more columns/custom filters to cross. Clicking this generates every combination of their values (e.g., 3 shifts × 7 days = up to 21 groups) and adds each one as a group. Combinations with no matching data and those that are duplicates are skipped automatically. A warning will be shown before proceeding if a combination exceeds 300 groups.
Delete All Groups	Removes all existing custom analysis groups.

Once created, groups appear as selectable rows/columns anywhere the Row or Column Source is set to 'Custom Analysis Groups'.

Running the Test & Reading Results

Click Run Analysis at the bottom once you have selected all desired statistical test settings. The statistical test that runs depends on your setup:

Feature/Button	What it does
1D (Goodness-of-Fit)	Incident Count When Columns = None. A chi-square test comparing each Row's observed count to the count expected by a uniform distribution.
2D (Test of Independence)	Incident Count When Columns ≠ None. A chi-square test of independence across a Rows x Columns table. If any expected cell count is below 5 Fisher's Exact Test is used instead.
Efficiency (One-Sample Z/T-test)	When Metric = Efficiency. For each selected Row, compares its mean efficiency (distance ÷ timeElapsed) against the population mean of all <u>filtered</u> incidents, using a Z-test when there are ≥30 qualifying incidents and a T-test otherwise.

The results panel opens with a summary card capturing the test used, the metric (incident count vs efficiency), any active Data/Custom filters that were used, the test statistic and p-value, and a significance rating (Not/Marginally/Significant/Highly Significant).

There is also an expandable Full Results Table that shows all the observed/expected counts (or group means for efficiency).

NOTE: The statistical tests run entirely on filtered data, so anything checked in Data Filters or Custom Filters on the map narrows the population before the test even starts.

8. Import/Export

Export All Settings and Import All Settings essentially allow you to “save” and “reload” your current session, respectively, including any areas and/or categories you may have drawn, custom filters, map coordinates, and station location are included.

NOTE: incident CSV rows are never included for data-privacy reasons.

NOTE: Custom Analysis Groups are not included in the export file. If you rely on them, note down your group definitions separately, or re-create them with the Auto-Generate Combinations options after importing the other saved settings on a new device.

Export Files → Areas & Categories allows you to export only the coordinates of the vertices of the drawn polygons, allowing you to “save” your region designations to, for example, load a new incident data file on another device without having to draw each area and category again. This does not save any other session information.

NOTE: Whenever importing a data file, even if to the same browser, a distance (re)calculation may be necessary. These may take a few minutes depending on the size of the data file.

9. Keyboard Shortcuts

Feature/Button	What it does
Ctrl+Z/Cmd+Z (while drawing)	Undo the last placed vertex. Works for areas and categories.
Right click on a vertex (while drawing), inside a finished polygon, or the station marker	Deletes the vertex, polygon, or station marker.
Enter (while a popup is open)	Confirms/saves the current settings displayed on the popup. This applies across Custom Filter creation, Category Naming, Editing Locations, Analysis Group creation, Auto-Generate Combinations option, and the area/category Edit toolbar.
Esc (while any popup or edit mode is open)	Cancels/closes that mode/popup without saving
Click and drag, and scroll on the map	Pan and zoom, respectively.
Click inside a finished polygon	Opens polygon options: edit vertices, rename, edit area selections for categories, etc. Be sure to save after all desired changes are made.